

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
Written: 2005-12-09 06:50:52 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:09 (EST)

Sectors:
313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328
329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344
345 346 347 348 349 350 351 352
353 354 355 356 357 358

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ blkcat -o 0 Ch01InChap01.dd 237
*****
*****
*****
*****
*****

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ blkcat -o 0 Ch01InChap01.dd 284
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business
```

Home kali-linux-2025.3-vmware... 1 2 3 4

kali@kali: ~/Forensics_Project_Lab3/evidence

Session Actions Edit View Help

password: 342rroi9

Thank you for your business

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
\$ blkcat -o 0 Ch01InChap01.dd 312

Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
\$ blkcat -o 0 Ch01InChap01.dd 313

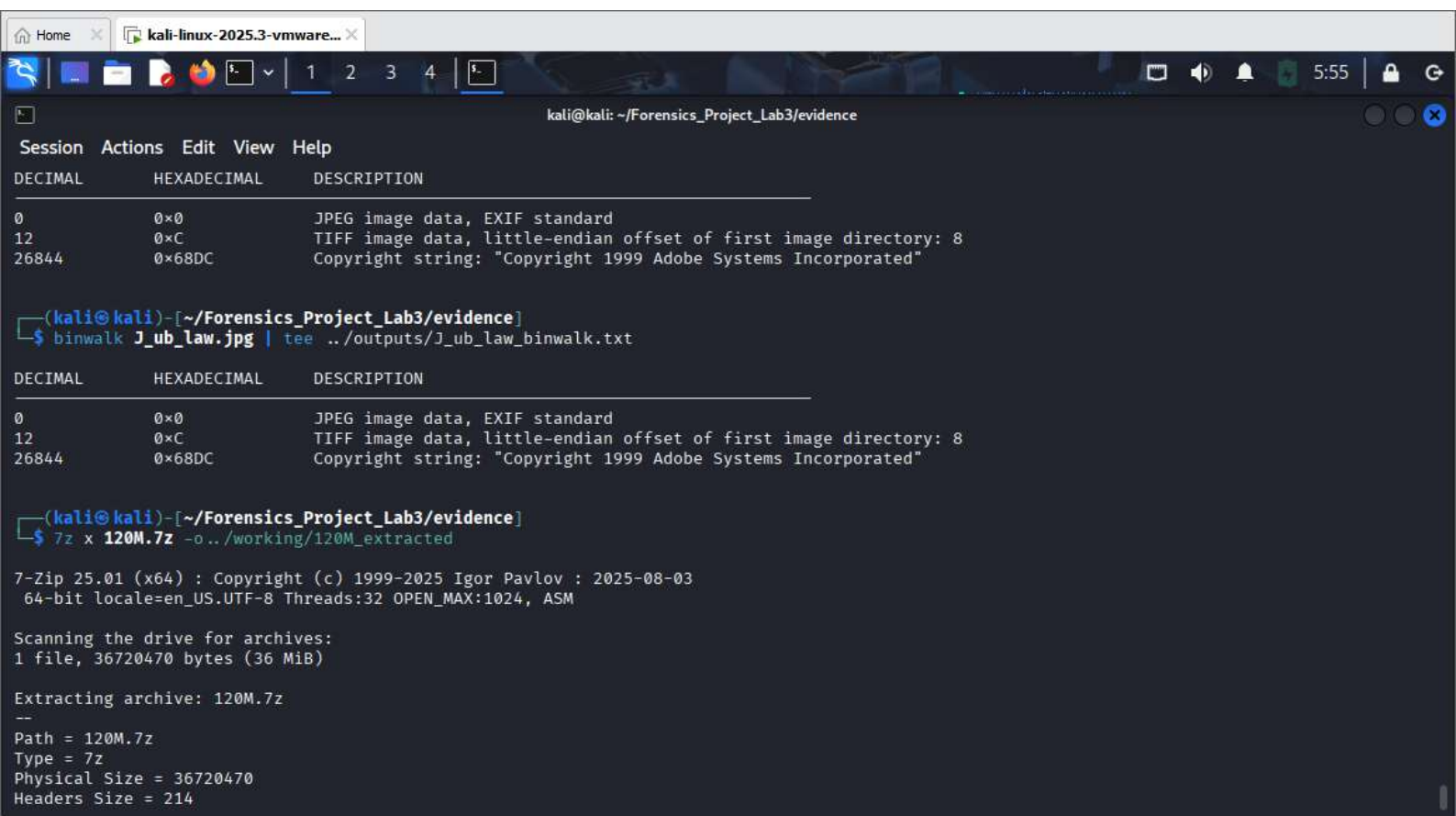
.....
.....
.....
.....

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
\$ which binwalk

/usr/bin/binwalk

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
\$ binwalk J_ub_law.jpg

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8



```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size: 124782229
Compressed: 36720470

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ file ../working/120M_extracted/*
../working/120M_extracted/120M: directory

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum ../working/USB_IMAGE
sha256sum ../working/USB_IMAGE
md5sum: ../working/USB_IMAGE: No such file or directory
sha256sum: ../working/USB_IMAGE: No such file or directory

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ ls -lah ../working
total 5.0M
drwxrwxr-x 3 kali kali 4.0K Sep  3 18:17 .
drwxrwxr-x 10 kali kali 4.0K Sep  3 10:28 ..
drwxrwxr-x 3 kali kali 4.0K Sep  3 18:17 120M_extracted
-rw-rw-r-- 1 kali kali 3.3M Sep  3 13:42 J_ub_law.hex
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence

Session Actions Edit View Help

drwxrwxr-x 3 kali kali 4.0K Sep 3 18:17 .
drwxrwxr-x 10 kali kali 4.0K Sep 3 10:28 ..
drwxrwxr-x 3 kali kali 4.0K Sep 3 18:17 120M_extracted
-rw-rw-r-- 1 kali kali 3.3M Sep 3 13:42 J_ub_law.hex
-rw-rw-r-- 1 kali kali 1.7M Sep 3 13:44 J_ub_law_reconstructed.jpg

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ find ../working -maxdepth 3 -type f -printf '%p\n'
../working/J_ub_law_reconstructed.jpg
../working/120M_extracted/120M/usb_fat_carving.001
../working/120M_extracted/120M/usb_fat_carving.001.txt
../working/J_ub_law.hex

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ ls -lah ../working/120M_extracted/120M
total 120M
drwxrwxr-x 2 kali kali 4.0K Sep 22 2021 .
drwxrwxr-x 3 kali kali 4.0K Sep 3 18:17 ..
-rw-rw-r-- 1 kali kali 119M Sep 22 2021 usb_fat_carving.001
-rw-rw-r-- 1 kali kali 1.7K Sep 22 2021 usb_fat_carving.001.txt

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum ../working/120M_extracted/120M/usb_fat_carving.001
ba4a1d0ba49f4a6667b00a3b3e85e604 ../working/120M_extracted/120M/usb_fat_carving.001

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sha256sum ../working/120M_extracted/120M/usb_fat_carving.001
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 ../working/120M_extracted/120M/usb_fat_carving.001

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum ../working/120M_extracted/120M/usb_fat_carving.001 | tee ../hashes/usb_fat_carving_MD5.txt
ba4a1d0ba49f4a6667b00a3b3e85e604 ../working/120M_extracted/120M/usb_fat_carving.001
```

```
kali-linux-2025.3-vmware... X
1 2 3 4 | $
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum ../working/120M_extracted/120M/usb_fat_carving.001 | tee ../hashes/usb_fat_carving_MD5.txt
ba4a1d0ba49f4a6667b00a3b3e85e604 ../working/120M_extracted/120M/usb_fat_carving.001

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sha256sum ../working/120M_extracted/120M/usb_fat_carving.001 | tee ../hashes/usb_fat_carving_SHA256.txt
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 ../working/120M_extracted/120M/usb_fat_carving.001

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat ../hashes/usb_fat_carving_MD5.txt
cat ../hashes/usb_fat_carving_SHA256.txt
ba4a1d0ba49f4a6667b00a3b3e85e604 ../working/120M_extracted/120M/usb_fat_carving.001
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 ../working/120M_extracted/120M/usb_fat_carving.001

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ img_stat ../working/120M_extracted/120M/usb_fat_carving.001
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size: 512

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ mmls ../working/120M_extracted/120M/usb_fat_carving.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length      Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
```

```
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ mkdir -p ../recovered/scalpel_output_2

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sudo scalpel ../working/120M_extracted/120M/usb_fat_carving.001 -o ../recovered/scalpel_output_2
[sudo] password for kali:
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

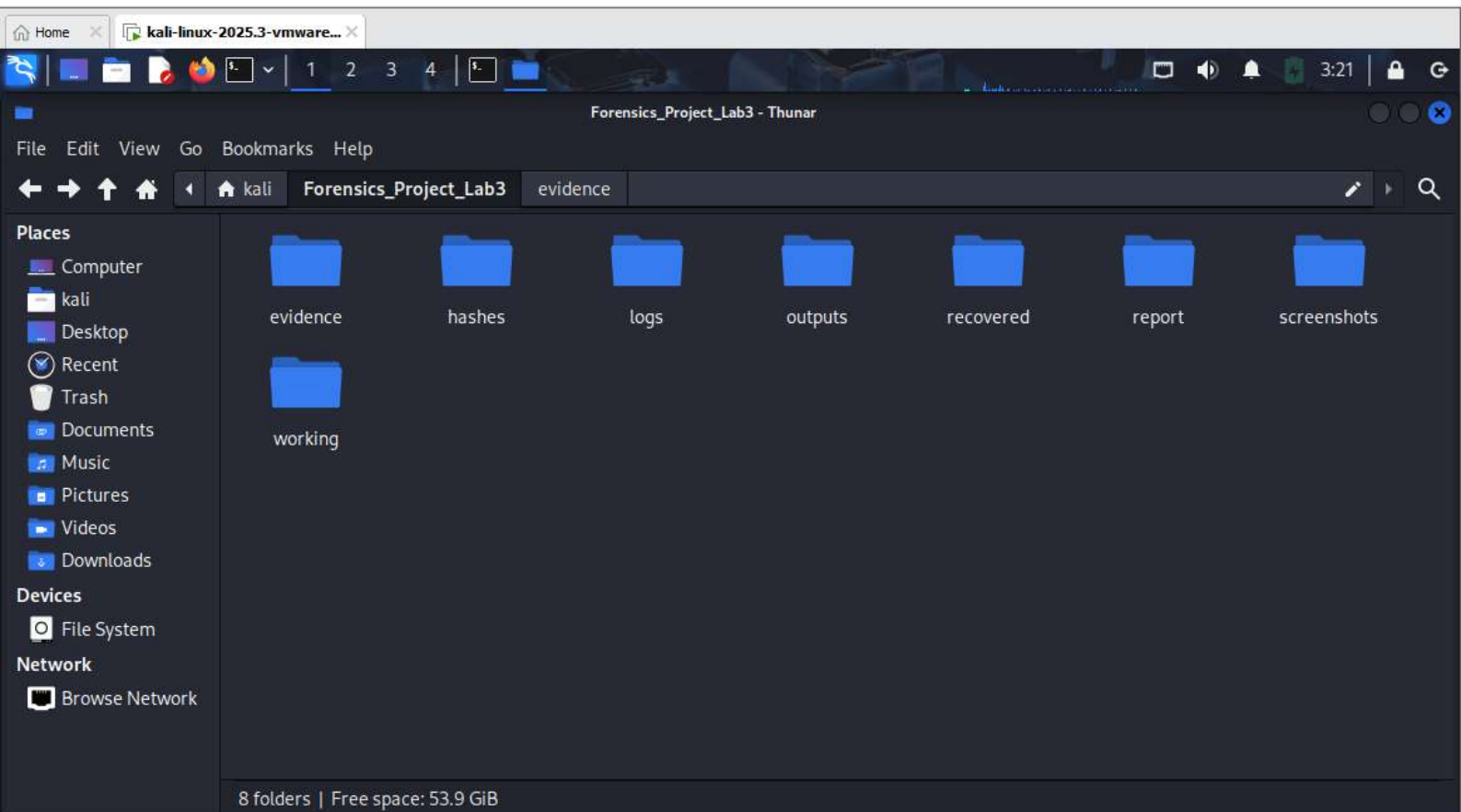
Opening target "/home/kali/Forensics_Project_Lab3/working/120M_extracted/120M/usb_fat_carving.001"
Image file pass 1/2.
../working/120M_extracted/120M/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues ...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" → 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" → 9 files
Carving files from image.
Image file pass 2/2.
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
../working/120M_extracted/120M_usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 45 seconds.

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ find ../recovered/scalpel_output_2 -type f
../recovered/scalpel_output_2/jpg-1-0/00000009.jpg
../recovered/scalpel_output_2/jpg-1-0/00000010.jpg
../recovered/scalpel_output_2/jpg-1-0/00000014.jpg
../recovered/scalpel_output_2/jpg-1-0/00000011.jpg
../recovered/scalpel_output_2/jpg-1-0/00000012.jpg
../recovered/scalpel_output_2/jpg-1-0/00000008.jpg
../recovered/scalpel_output_2/jpg-1-0/00000016.jpg
../recovered/scalpel_output_2/jpg-1-0/00000013.jpg
../recovered/scalpel_output_2/jpg-1-0/00000015.jpg
../recovered/scalpel_output_2/jpg-0-0/00000000.jpg
../recovered/scalpel_output_2/jpg-0-0/00000004.jpg
../recovered/scalpel_output_2/jpg-0-0/00000006.jpg
../recovered/scalpel_output_2/jpg-0-0/00000003.jpg
../recovered/scalpel_output_2/jpg-0-0/00000005.jpg
../recovered/scalpel_output_2/jpg-0-0/00000002.jpg
../recovered/scalpel_output_2/jpg-0-0/00000007.jpg
../recovered/scalpel_output_2/jpg-0-0/00000001.jpg
../recovered/scalpel_output_2/audit.txt

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ find ../recovered/scalpel_output_2 -type f | wc -l
18

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$
```



```
Home kali-linux-2025.3-vmware...
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ ls -lh J_ub_law.jpg
-rwxrw-rw- 1 kali kali 1.7M Sep  3 08:48 J_ub_law.jpg
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ stat J_ub_law.jpg
File: J_ub_law.jpg
Size: 1692150      Blocks: 3312      IO Block: 4096   regular file
Device: 8,1      Inode: 3548879    Links: 1
Access: (0766/-rwxrw-rw-)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-09-03 10:34:24.734249615 -0400
Modify: 2026-09-03 08:48:24.029515000 -0400
Change: 2026-09-03 10:34:24.435782398 -0400
Birth: 2026-09-03 10:34:24.412219196 -0400
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum J_ub_law.jpg | tee ../hashes/J_ub_law_MD5.txt
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sha256sum J_ub_law.jpg | tee ../hashes/J_ub_law_SHA256.txt
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cd ~/Forensics_Project_Lab3/evidence
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
└─$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
└─$ xxd -l 32 J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 ...Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 ...@.

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
└─$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae ..=R..0....." ..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7 ~.b.z...z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981 .U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9 U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 ....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 .*.-$.....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8...o..Z.T.e
0019d1f0: d351 9ebf ffd9 ..Q....

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
└─$ xxd -p J_ub_law.jpg > ../working/J_ub_law.hex
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
0019d1f0: d351 9ebf ffd9 .Q....

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ xxd -p J_ub_law.jpg > ../working/J_ub_law.hex

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ head ../working/J_ub_law.hex
ffd8ffe117f045786966000049492a000800000010000001030001000000
401300000101030001000000d00c00000201030003000000ce000000601
03000100000002000000f0102001200000d4000000100102000900000
e60000001201030001000000010000001501030001000000030000001a01
050001000000ef0000001b01050001000000f70000002801030001000000
020000003101020021000000ff0000003201020014000000200100009882
02000c000000340100009e9c010052000000400100006987040001000000
9401000044040000800080008004e494b4f4e20434f52504f524154494f
4e004e494b4f4e20443400c0c62d0010270000c0c62d001027000041646f
62652050686f746f73686f702032312e3120284d6163696e746f73682900

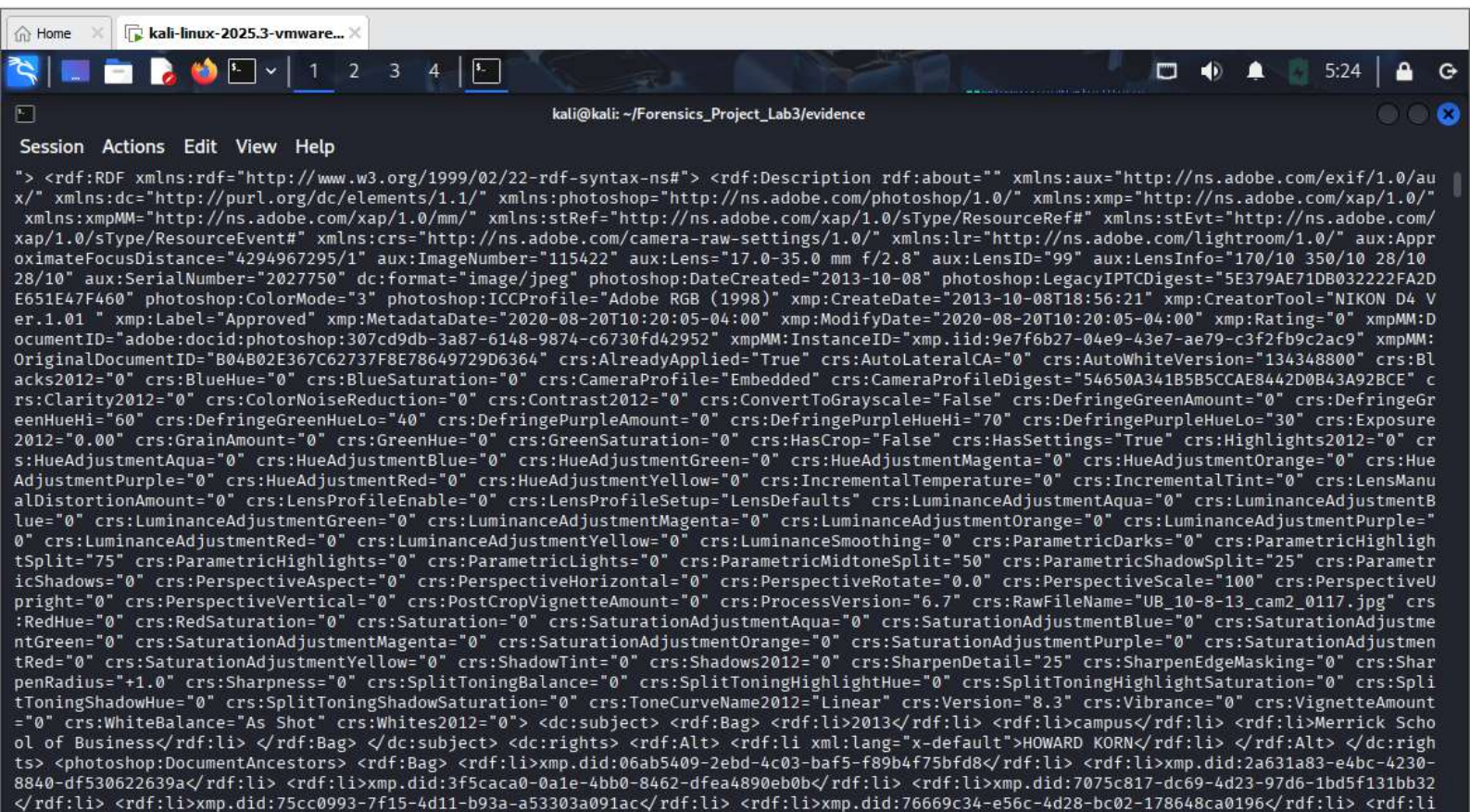
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ xxd -r -p ../working/J_ub_law.hex ../working/J_ub_law_reconstructed.jpg

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ ls -lh ../working/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  3 13:44 ../working/J_ub_law_reconstructed.jpg

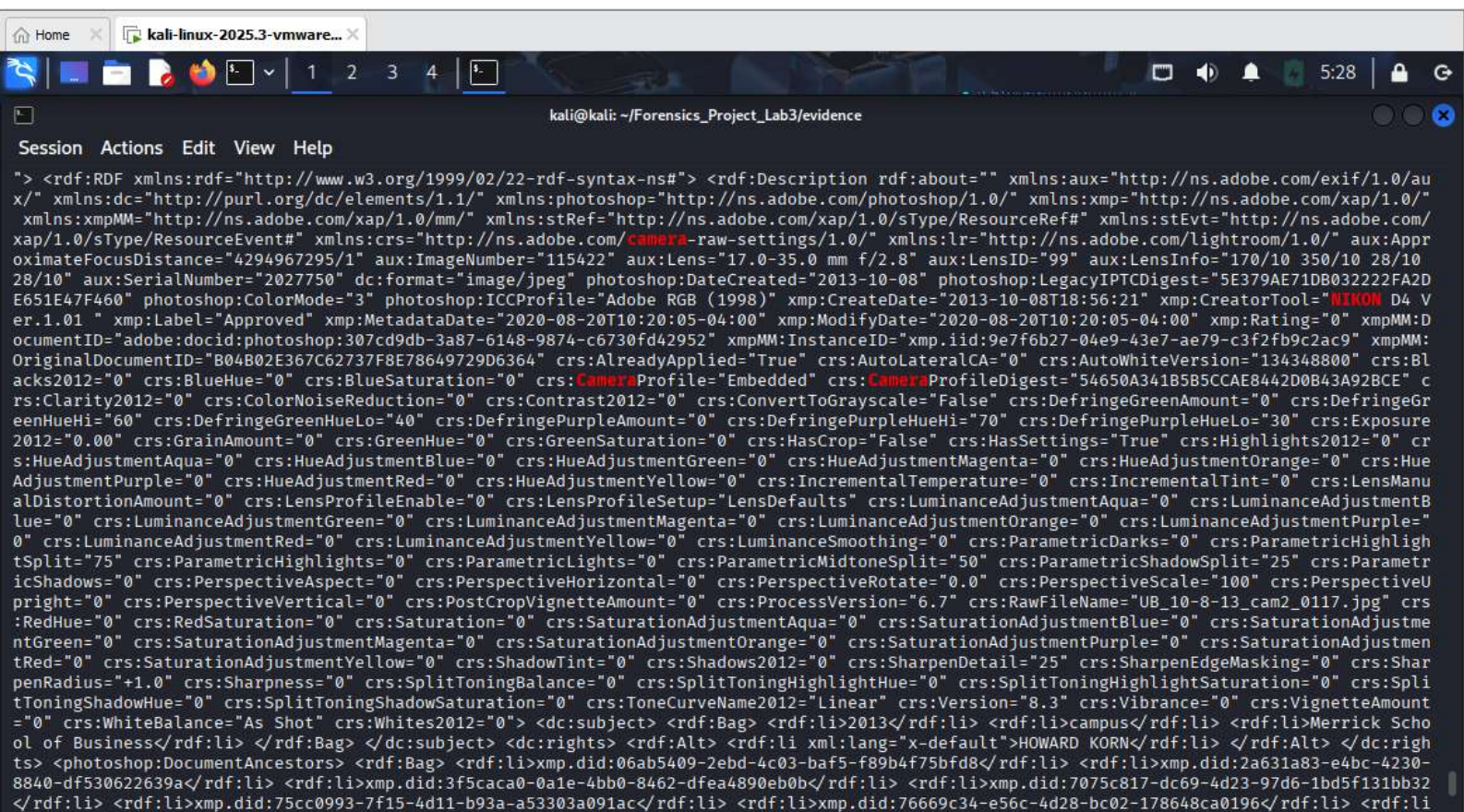
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum ../working/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  ../working/J_ub_law_reconstructed.jpg

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sha256sum ../working/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  ../working/J_ub_law_reconstructed.jpg
```

```
Home kali-linux-2025.3-vmware... X
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ sha256sum ../working/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  ../working/J_ub_law_reconstructed.jpg
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ md5sum J_ub_law.jpg
sha256sum J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ strings -a -n 4 J_ub_law.jpg | tee ../outputs/J_ub_law_strings.txt
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r
7GWgw
AQaq"
dEU6te
'7GWgw
2VtWX
```



```
kali-linux-2025.3-vmware... X
Home
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
X{)6,
~}'j6
l.~k,
t':k
K_WK%
:r{;Y
woMO=eM8
Evbu
4mC#
<so~
qp0#
]>1X
x#-V
3u!P[C
vdW
H}Lt
20E(>)
?4|)
RT 8
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ less ../outputs/J_ub_law_strings.txt
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ strings -a J_ub_law.jpg | grep -Ei 'make|model|camera|canon|nikon|sony|iphone|samsung'
NIKON CORPORATION
NIKON D4
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38
"> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/au
x/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/"
xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/
xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:Appr
```



```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help

<?xpacket end="w"?>

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ strings -a J_ub_law.jpg | grep -Ei 'software|photoshop|lightroom|adobe|gimp'
Adobe Photoshop 21.1 (Macintosh)
Adobe_CM
Adobe
"Photoshop 3.0
Adobe_CM
Adobe
http://ns.adobe.com/xap/1.0/
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38
"> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/au
x/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/"
xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/
xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:Appr
oximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10
28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08" photoshop:LegacyIPTCDigest="5E379AE71DB032222FA2D
E651E47F460" photoshop:ColorMode="3" photoshop:ICCPProfile="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 V
er.1.01" xmp:Label="Approved" xmp:MetadataDate="2020-08-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:D
ocumentID="adobe:docid:photoshop:307cd9db-3a87-6148-9874-c6730fd42952" xmpMM:InstanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" xmpMM:
OriginalDocumentID="B04B02E367C62737F8E78649729D6364" crs:AlreadyApplied="True" crs:AutoLateralCA="0" crs:AutoWhiteVersion="134348800" crs:Bl
acks2012="0" crs:BlueHue="0" crs:BlueSaturation="0" crs:CameraProfile="Embedded" crs:CameraProfileDigest="54650A341B5B5CCAE8442D0B43A92BCE" c
rs:Clarity2012="0" crs:ColorNoiseReduction="0" crs:Contrast2012="0" crs:ConvertToGrayscale="False" crs:DefringeGreenAmount="0" crs:DefringeGr
eenHueHi="60" crs:DefringeGreenHueLo="40" crs:DefringePurpleAmount="0" crs:DefringePurpleHueHi="70" crs:DefringePurpleHueLo="30" crs:Exposure
2012="0.00" crs:GrainAmount="0" crs:GreenHue="0" crs:GreenSaturation="0" crs:HasCrop="False" crs:HasSettings="True" crs:Highlights2012="0" cr
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
Gimp

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ strings -a J_ub_law.jpg | grep -Ei 'date|time|202[0-9]|201[0-9]'
2020:08:20 10:20:05
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
20131008
2013
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38
"> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/au
x/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/"
xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/
xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:Appr
oximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10
28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DataCreated="2013-10-08" photoshop:LegacyIPTCDigest="5E379AE71DB032222FA2D
E651E47F460" photoshop:ColorMode="3" photoshop:ICCProfile="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 V
er.1.01" xmp:Label="Approved" xmp:MetadataDate="2020-08-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:D
ocumentID="adobe:docid:photoshop:307cd9db-3a87-6148-9874-c6730fd42952" xmpMM:InstanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" xmpMM:
OriginalDocumentID="B04B02E367C62737F8E78649729D6364" crs:AlreadyApplied="True" crs:AutoLateralCA="0" crs:AutoWhiteVersion="134348800" crs:Bl
acks2012="0" crs:BlueHue="0" crs:BlueSaturation="0" crs:CameraProfile="Embedded" crs:CameraProfileDigest="54650A341B5B5CCAE8442D0B43A92BCE" c
rs:Clarity2012="0" crs:ColorNoiseReduction="0" crs:Contrast2012="0" crs:ConvertToGrayscale="False" crs:DefringeGreenAmount="0" crs:DefringeGr
eenHueHi="60" crs:DefringeGreenHueLo="40" crs:DefringePurpleAmount="0" crs:DefringePurpleHueHi="70" crs:DefringePurpleHueLo="30" crs:Exposure
2012="0.00" crs:GrainAmount="0" crs:GreenHue="0" crs:GreenSaturation="0" crs:HasCrop="False" crs:HasSettings="True" crs:Highlights2012="0" cr
s:HueAdjustmentAqua="0" crs:HueAdjustmentBlue="0" crs:HueAdjustmentGreen="0" crs:HueAdjustmentMagenta="0" crs:HueAdjustmentOrange="0" crs:Hue
AdjustmentPurple="0" crs:HueAdjustmentRed="0" crs:HueAdjustmentYellow="0" crs:IncrementalTemperature="0" crs:IncrementalTint="0" crs:LensManu
alDistortionAmount="0" crs:LensProfileEnable="0" crs:LensProfileSetup="LensDefaults" crs:LuminanceAdjustmentAqua="0" crs:LuminanceAdjustmentB
lue="0" crs:LuminanceAdjustmentGreen="0" crs:LuminanceAdjustmentMagenta="0" crs:LuminanceAdjustmentOrange="0" crs:LuminanceAdjustmentPurple="
0" crs:LuminanceAdjustmentRed="0" crs:LuminanceAdjustmentYellow="0" crs:LuminanceSmoothing="0" crs:ParametricDarks="0" crs:ParametricHighligh
tSplit="75" crs:ParametricHighlights="0" crs:ParametricLights="0" crs:ParametricMidtoneSplit="50" crs:ParametricShadowSplit="25" crs:Parametr
icShadows="0" crs:PerspectiveAspect="0" crs:PerspectiveHorizontal="0" crs:PerspectiveRotate="0.0" crs:PerspectiveScale="100" crs:PerspectiveU
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence

<?xpacket end="w"?>

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ strings -a J_ub_law.jpg | grep -Ei 'gps|latitude|longitude|location'
PgPsenum
PgPs

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ mmls Ch01InChap01.dd

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
```

Home kali-linux-2025.3-vmware...

kali@kali: ~/Forensics_Project_Lab3/evidence

Session Actions Edit View Help

OEM Name: ,'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: ♦3♦♦M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION

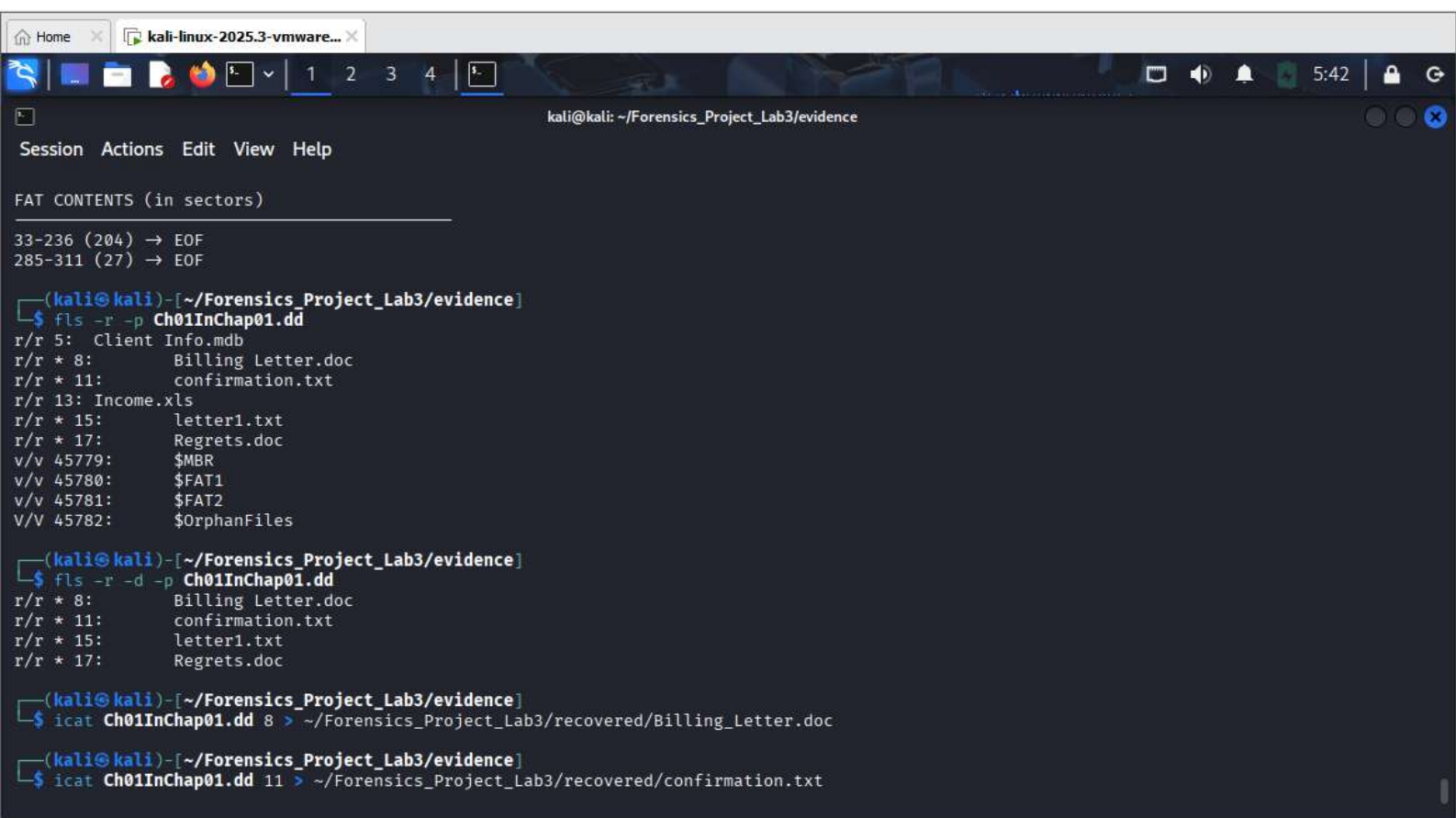
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)

33-236 (204) → EOF



The screenshot shows a Kali Linux terminal window with the title bar "kali-linux-2025.3-vmware...". The terminal is running in the directory `~/Forensics_Project_Lab3/evidence`. The output shows the results of a file recovery process using the `icat` tool. The first command, `icat Ch01InChap01.dd`, lists all recovered files and their locations. The second command, `icat Ch01InChap01.dd 8`, shows the location of the file at sector 8. The third command, `icat Ch01InChap01.dd 11`, shows the location of the file at sector 11.

```
kali@kali: ~/Forensics_Project_Lab3/evidence

FAT CONTENTS (in sectors)
-----
33-236 (204) → EOF
285-311 (27) → EOF

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat Ch01InChap01.dd 8 > ~/Forensics_Project_Lab3/recovered/Billing_Letter.doc

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat Ch01InChap01.dd 11 > ~/Forensics_Project_Lab3/recovered/confirmation.txt
```

The screenshot shows a Kali Linux terminal window titled "kali@kali: ~/Forensics_Project_Lab3/evidence". The terminal displays a series of commands and their outputs:

```
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ icat Ch01InChap01.dd 11 > ~/Forensics_Project_Lab3/recovered/confirmation.txt

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ icat Ch01InChap01.dd 15 > ~/Forensics_Project_Lab3/recovered/letter1.txt

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ icat Ch01InChap01.dd 17 > ~/Forensics_Project_Lab3/recovered/Regrets.doc

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ ls -lh ~/Forensics_Project_Lab3/recovered
total 56K
-rw-rw-r-- 1 kali kali 24K Sep  3 15:49 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Sep  3 15:49 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Sep  3 15:49 letter1.txt
-rw-rw-r-- 1 kali kali 23K Sep  3 15:50 Regrets.doc

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ file ~/Forensics_Project_Lab3/recovered/*
/home/kali/Forensics_Project_Lab3/recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10
, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2,
Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
/home/kali/Forensics_Project_Lab3/recovered/confirmation.txt: ASCII text, with CRLF line terminators
/home/kali/Forensics_Project_Lab3/recovered/letter1.txt: ASCII text, with CRLF line terminators
/home/kali/Forensics_Project_Lab3/recovered/Regrets.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10
, Code page: 1252, Title: 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2
, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
```

The screenshot shows a Kali Linux desktop environment. At the top, there is a taskbar with icons for Home, a file manager, and a terminal window titled 'kali-linux-2025.3-vmware...'. Below the taskbar is a panel with application icons including Firefox, a file manager, and a terminal. The terminal window is open, displaying the following content:

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat ~/Forensics_Project_Lab3/recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

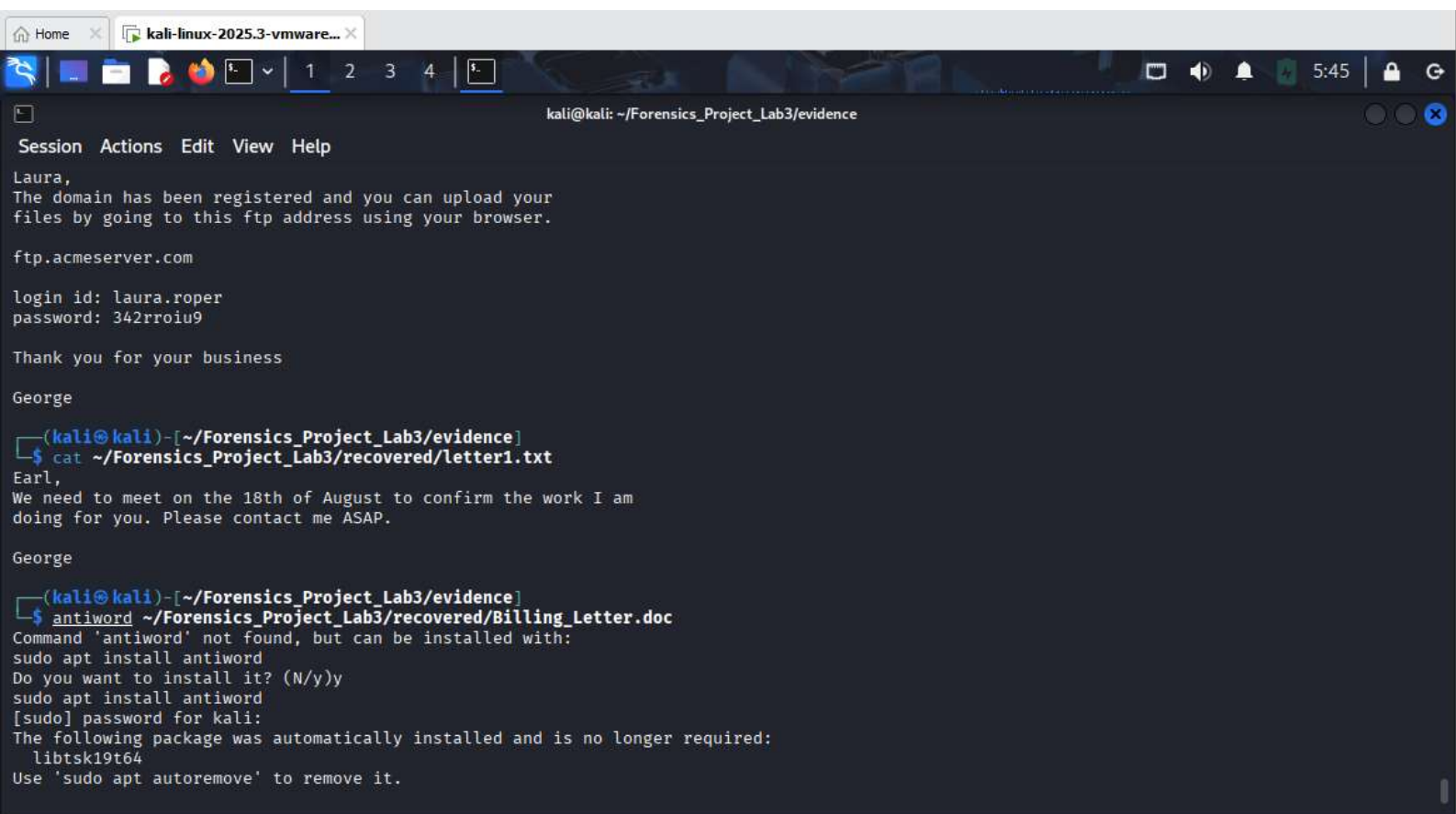
Thank you for your business

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat ~/Forensics_Project_Lab3/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat ~/Forensics_Project_Lab3/recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.
```



```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ cat ~/Forensics_Project_Lab3/recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ antiword ~/Forensics_Project_Lab3/recovered/Billing_Letter.doc
Command 'antiword' not found, but can be installed with:
sudo apt install antiword
Do you want to install it? (N/y)y
sudo apt install antiword
[sudo] password for kali:
The following package was automatically installed and is no longer required:
libtss19t64
Use 'sudo apt autoremove' to remove it.
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help
The following package was automatically installed and is no longer required:
  libtsk19t64
Use 'sudo apt autoremove' to remove it.

Installing:
  antiword

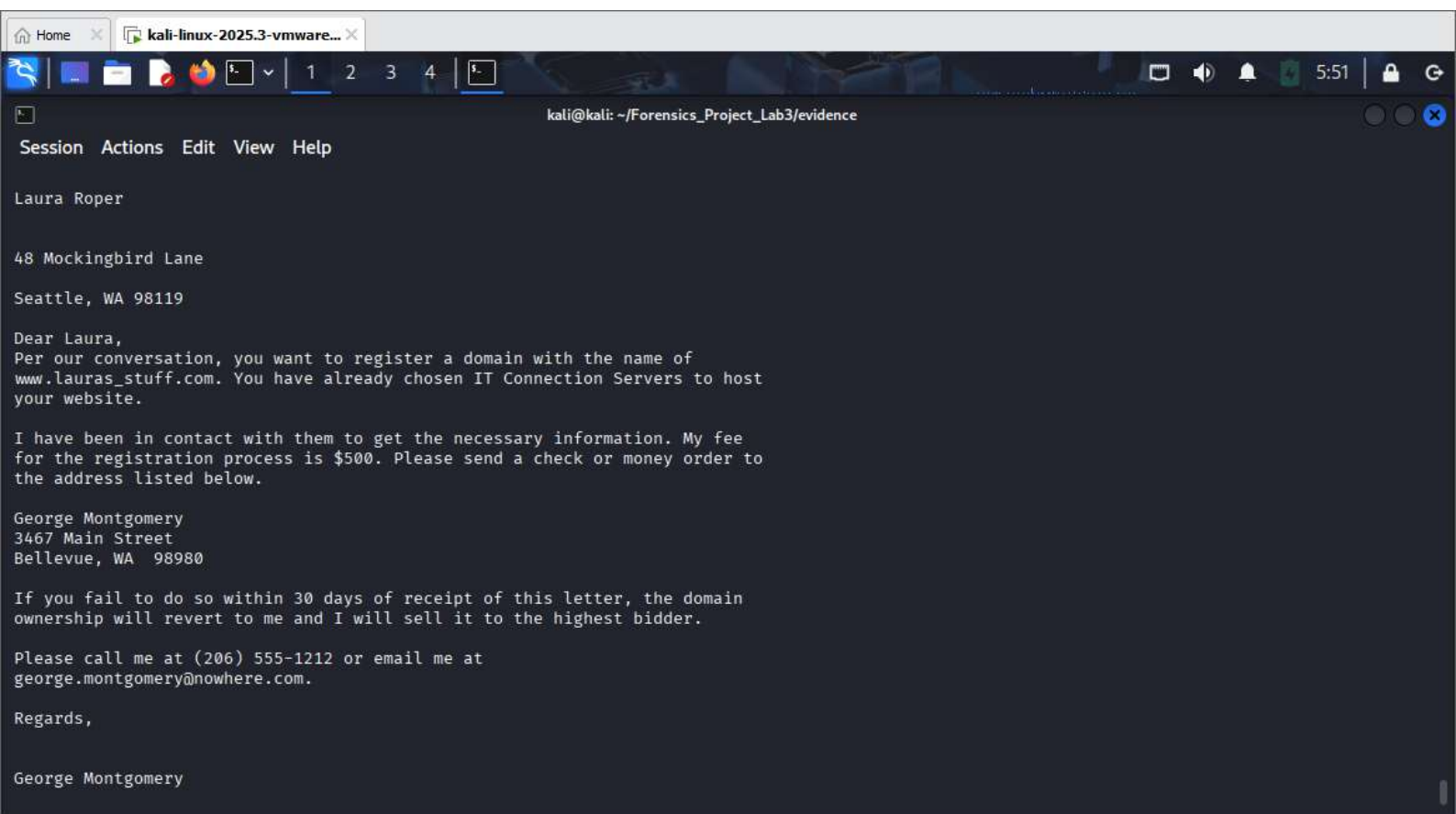
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2309
  Download size: 123 kB
  Space needed: 601 kB / 58.1 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 antiword amd64 0.37-17 [123 kB]
Fetched 123 kB in 9s (13.8 kB/s)
Selecting previously unselected package antiword.
(Reading database ... 417872 files and directories currently installed.)
Preparing to unpack .../antiword_0.37-17_amd64.deb ...
Unpacking antiword (0.37-17) ...
Setting up antiword (0.37-17) ...
Processing triggers for kali-menu (2025.3.2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ antiword ~/Forensics_Project_Lab3/recovered/Billing_Letter.doc

13 October 2005

Laura Roper
```



```
kali@kali: ~/Forensics_Project_Lab3/evidence
Session Actions Edit View Help

George Montgomery

(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ antiword ~/Forensics_Project_Lab3/recovered/Regrets.doc

02 November 2005

Randall Watson
89 Darnell Street
Des Moines, WA 98000

Dear Mr. Watson,
I regret to inform you that all five versions of the domain name that you
chose have already been purchased. If you have other variations, please let
me know by calling me at (206) 555-1212 or email me at
george.montgomery@nowhere.com

Regards,

George Montgomery

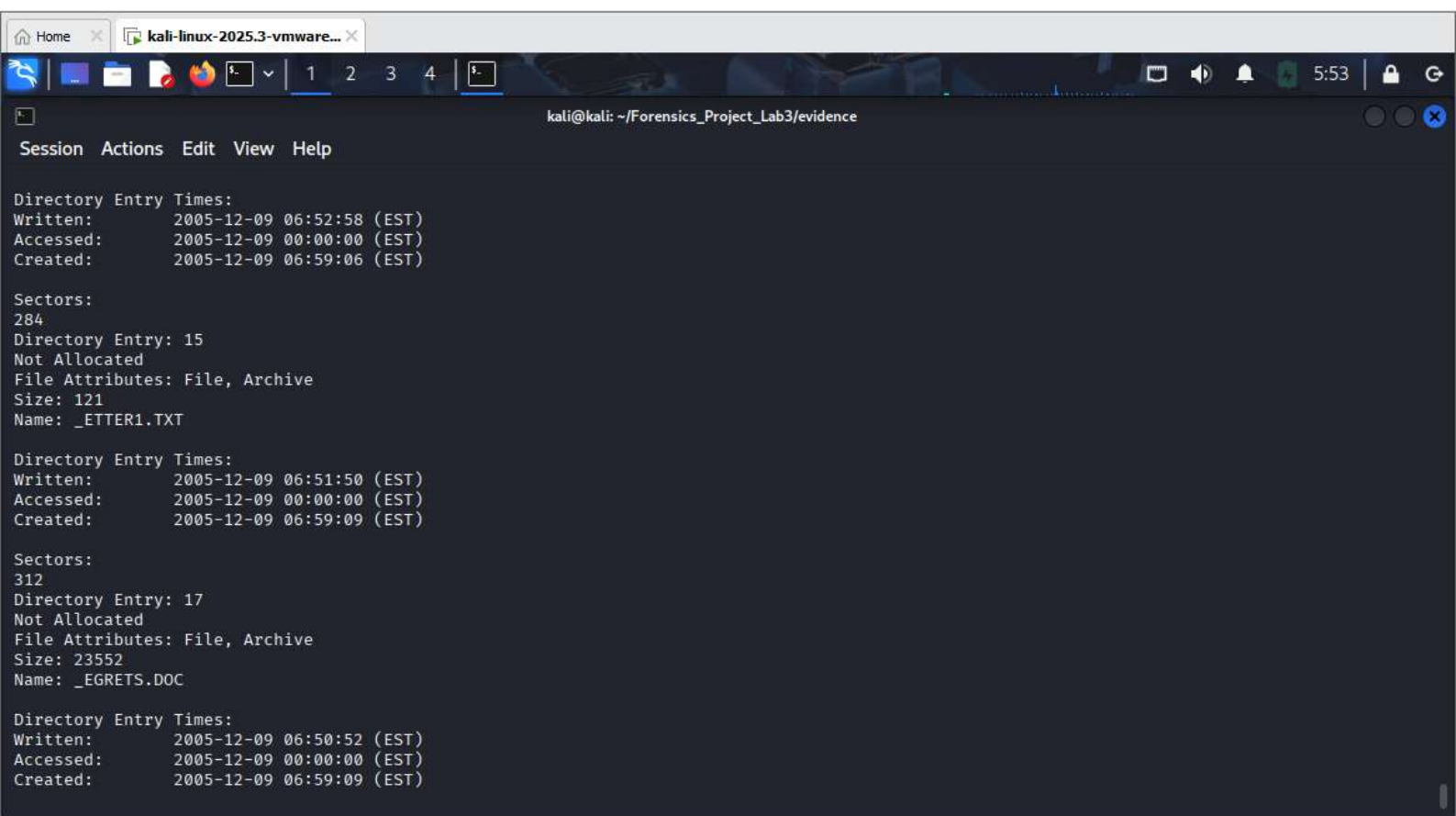
(kali@kali)-[~/Forensics_Project_Lab3/evidence]
$ istat Ch01InChap01.dd 8
istat Ch01InChap01.dd 11
istat Ch01InChap01.dd 15
```

```
kali@kali: ~/Forensics_Project_Lab3/evidence
$ istat Ch01InChap01.dd 8
istat Ch01InChap01.dd 11
istat Ch01InChap01.dd 15
istat Ch01InChap01.dd 17
Directory Entry: 8
Not Allocated
File Attributes: File, Archive
Size: 24064
Name: _ILLIN~1.DOC

Directory Entry Times:
Written: 2005-12-09 06:50:28 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:05 (EST)

Sectors:
237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written: 2005-12-09 06:52:58 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
```



Data Carving with XXD, Binwalk and Scalpel Report

Student Name: Michael Uke

Environment: Kali Linux

Primary Evidence: Ch01InChap01.dd

Additional Evidence: J_ub_law.jpg; 120M.7z / usb_fat_carving.001

Tools: The Sleuth Kit, xxd, strings, file, antiword, MD5/SHA-256, Binwalk and Scalpel

Date: September 2026

1. Introduction

This laboratory involved forensic examination of digital evidence using Kali Linux. The work covered evidence preservation and hashing, JPEG hexadecimal analysis, strings and metadata examination, filesystem identification, deleted-file discovery and recovery, sector level analysis, USB evidence identification, and preparation for JPEG file carving. Original evidence was kept separate from working and recovery outputs.

2. Forensic Workspace

A dedicated workspace was created at ~/Forensics_Project_Lab3 with separate evidence, hashes, working, recovered, outputs, screenshots, logs and report areas. This separation helped preserve original evidence and distinguish recovered material from source evidence.

3. JPEG Evidence J_ub_law.jpg

The JPEG evidence was examined without altering the original. The recorded file size was 581,331 bytes. MD5: 2e38edb92662bd14a6eca813bfc5572f. SHA-256: 0d16ea7c5523c68b57abb568cbb91fd116f8bdb5830b863bb2ea2acafcfd7e4. The image was identified as JPEG/JFIF, 1920 × 1080.

- xxd confirmed the JPEG Start of Image marker FF D8 and JFIF information.
- The file ended with FF D9, the JPEG End of Image marker.
- A plain hexadecimal dump was created and reversed to reconstruct J_ub_law_reconstructed.jpg.
- strings revealed JFIF and ICC colour-profile strings including ICC_PROFILE, acsp, rXYZ, gXYZ, bXYZ and related profile data.

- No readable camera make/model, software, GPS/location, person/name or clear date/time information was identified in the targeted strings.

4. Main Forensic Image Ch01InChap01.dd

img_stat identified Ch01InChap01.dd as a raw forensic image. The image size was 1,474,560 bytes and the sector size was 512 bytes.

Property	Result
Image type	Raw
Image size	1,474,560 bytes
Sector size	512 bytes
Filesystem	FAT12
Filesystem start	Sector 0
Filesystem range	Sectors 0–2879

mmls did not return partition-table information. fsstat subsequently established that the image itself contained a FAT12 filesystem beginning at sector 0, so no additional partition offset was required.

5. Deleted Files and Recovery

Recursive fls analysis identified four deleted files. Each was recovered with icat and stored in the separate recovered directory.

File	Entry/Inode	Status	Sectors / Size
Billing Letter.doc	8	Deleted	237–283 / 24,064 bytes
confirmation.txt	11	Deleted	284 / 227 bytes
letter1.txt	15	Deleted	312 / 121 bytes
Regrets.doc	17	Deleted	313–358 / 23,552 bytes

file verified the recovered types: the two .doc files were Microsoft Composite Document File V2 files, while the two .txt files were ASCII text.

6. Recovered Evidence Findings

- confirmation.txt contained FTP related correspondence including a server hostname, username, password and upload instructions. Sensitive credentials were not reproduced in this report.
- letter1.txt contained a work-related message addressed to Earl requesting a meeting on 18 August.
- Billing Letter.doc contained correspondence dated 13 October 2005 concerning domain registration, website hosting, a \$500 registration fee and payment conditions.
- Regrets.doc contained correspondence dated 2 November 2005 concerning the purchase of five domain-name variations.

7. Metadata and Sector Level Analysis

istat confirmed that all four deleted entries were not allocated and provided their actual short names, sizes and sector locations. Because the FAT12 filesystem begins at sector 0, blkcat analysis used -o 0 and the actual sectors reported by istat.

- Billing Letter.doc: entry 8; sectors 237–283.
- confirmation.txt: entry 11; sector 284.
- letter1.txt: entry 15; sector 312.
- Regrets.doc: entry 17; sectors 313–358.

8. USB Evidence usb_fat_carving.001

During examination of the extracted 120M evidence, the actual USB carving evidence file was identified as usb_fat_carving.001 under working/120M_extracted/120M/. It was approximately 119 MB. An initial attempt to hash a placeholder name, USB_IMAGE, failed; directory inspection identified the correct filename.

- MD5: ba4a1d0ba49f4a6667b00a3b3e85e604
- SHA-256: 9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9

9. Scalpel File Carving Analysis

Scalpel 1.60 was available at /usr/bin/scalpel. The initial carving attempt failed because no file types were enabled. JPEG signatures for EXIF and JFIF were then enabled in scalpel.conf. A later attempt failed because the output directory was non empty, so a clean scalpel_output_2 directory was created. A final recovered file count was not recorded in the available evidence and is therefore not claimed.

10. Challenges Encountered

- mmls did not initially display a partition table; fsstat established a FAT12 filesystem beginning at sector 0.
- A placeholder USB_IMAGE filename caused an initial hashing error; the correct usb_fat_carving.001 file was then identified.
- Scalpel initially reported that no file types were configured; JPEG signatures were enabled.
- Scalpel rejected a non empty output directory; a clean output directory was created.
- A sudo authentication issue occurred during the Scalpel stage.

11. Forensic Integrity and Limitations

The investigation maintained separation between original evidence, working files and recovered material; cryptographic hashes were recorded; and actual filesystem, inode and sector values were obtained from forensic tool output rather than guessed. Sensitive credentials discovered in recovered evidence were not unnecessarily reproduced. The available record does not contain a final Scalpel recovery count, so no unsupported carving result is included.

12. Conclusion

The laboratory successfully demonstrated evidence preservation, hashing, JPEG structure analysis, strings examination, FAT12 filesystem identification, deleted file discovery and recovery, metadata and sector analysis, USB evidence identification and file carving preparation. Four deleted files were identified and recovered from Ch01InChap01.dd. The exercise also demonstrated the importance of verifying evidence filenames, using clean output directories, recording hashes, protecting sensitive information and relying on actual forensic tool output.